

Project: Decision Tree Classification Model

Part 2: ReturnWise Retail Analytics

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1. Business Overview

ReturnWise Retail Analytics is a decision-support project for a mid-sized e-commerce retailer selling apparel, footwear, electronics, beauty products, home goods, and accessories. The company receives thousands of online orders each month through its website, mobile app, marketplace partners, and social commerce channels. As online sales grow, product returns have become a major operational and financial challenge.

The sales and operations team must decide how to prioritize preventive actions because staff time and intervention budgets are limited. Some orders carry a high risk of being returned due to size fit problems, low product ratings, or long delivery times while others can proceed without special attention. A Decision Tree classification model helps the business identify patterns in order characteristics and customer behaviour, enabling smarter, earlier intervention decisions.

Element	Description
Business	ReturnWise Retail Analytics
Objective	Predict which orders are at high risk of being returned
Target Variable	Returned (Yes / No)
Dataset Size	455 records (450 after removing 5 duplicates)
Features	19 input features (10 numerical, 9 categorical)
Priority Metric	Recall and F1-Score: catch costly returns while limiting unnecessary interventions
Model Used	Decision Tree Classifier (sklearn, max_depth=4, random_state=42)

2. Data Preparation

2.1. Dataset Overview

The dataset contains 455 records representing individual customer orders. After removing 5 duplicate rows, 450 clean records were used. Each record has 22 original columns;

19 were retained as input features after dropping Order_ID, Purchase_Date, and the target variable Returned. Features fall into five business categories:

Feature Group	Columns
Customer Profile	Customer_Age, Customer_Segment, Region
Product Information	Product_Category, Product_Price, Product_Rating, Size_Fit_Issue_Level
Purchase Behaviour	Discount_Percent, Promotion_Type, Purchase_Channel, Payment_Method, Product_Page_Views
Order & Delivery	Shipping_Method, Delivery_Time_Days, Items_In_Order, Return_Window_Days
Customer History	Previous_Orders, Prior_Return_Count, Customer_Support_Contact

2.2. Class Distribution

The target variable Returned was imbalanced across the dataset, with non-returned orders forming the clear majority.

Class	Full Dataset (450)	Train / Test
Yes (Returned)	126 (28.0%)	101 / 27
No (Not Returned)	324 (72.0%)	259 / 63

2.3. Preprocessing Steps

- Dropped Order_ID (unique identifier with no predictive value) and Purchase_Date (date string; no time-series modelling applied)
- Removed 5 duplicate rows. The dataset reduced from 455 to 450 records
- Detected missing values across several columns. Numerical missing values filled with column median; categorical missing values filled with column mode (most frequent value)
- Confirmed zero missing values after imputation
- Applied One-Hot Encoding (sklearn ColumnTransformer + OneHotEncoder) to all 9 categorical features, expanding the feature space from 19 to 46 processed features
- Numerical features (10 columns) passed through unchanged, no scaling applied, as Decision Trees do not require feature scaling

- Split data 80/20 using train_test_split with random_state=42, producing 360 training records and 90 testing records.

3. Decision Tree Model

3.1. How Decision Trees Make Predictions

A Decision Tree is a supervised machine learning algorithm that recursively partitions data by asking binary questions about feature values. At each node, the algorithm selects the feature and threshold that best separates lead that became won deals from those that did not, using Gini impurity as the splitting criterion.

For ReturnWise Retail Analytics, the tree learns business rules such as: 'If Size_Fit_Issue_Level_High = 1 AND Prior_Return_Count > 2, predict Returned = Yes.' These rules are directly interpretable — an operations manager can follow any prediction path through the tree and understand exactly which factors the model responded to, which supports responsible deployment and human oversight.

3.2. Decision Tree Visualization

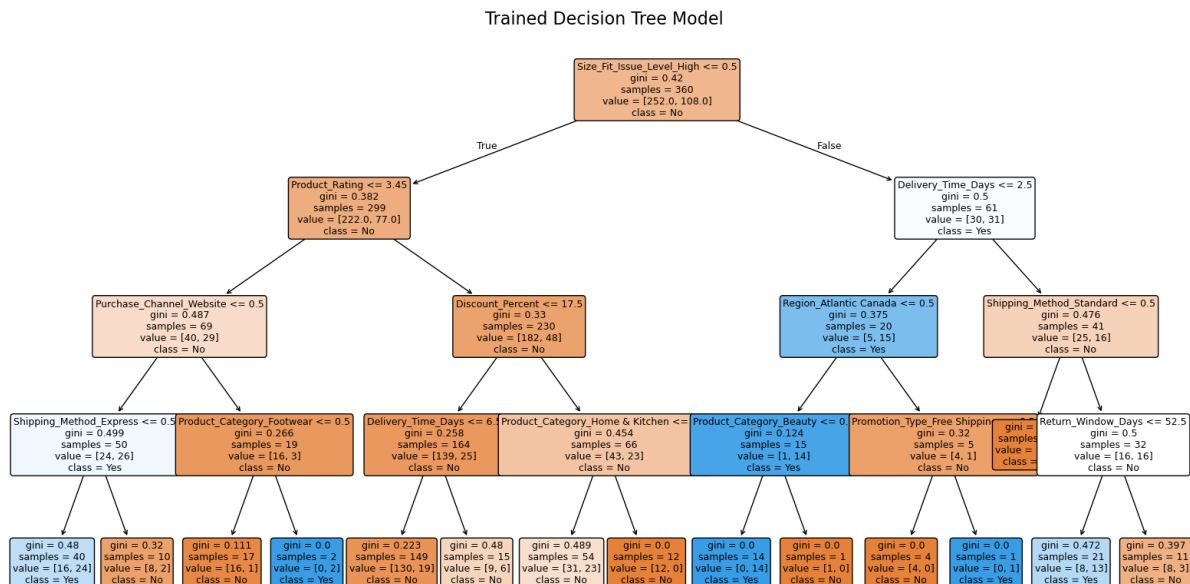


Figure 1: Decision Tree Model, ReturnWise Retail Analytics

The decision tree was trained using a maximum depth of 4, meaning it makes at most four sequential splitting decisions before reaching a final prediction. The tree is displayed with

colour-filled nodes, where the shading indicates the dominant class at each node darker shading toward one colour reflects greater class purity (a higher proportion of either Returned or Not Returned outcomes).

Starting from the root node at the top, the tree selects the single feature that best separates returned orders from non-returned ones. At each subsequent level, the tree continues subdividing the data using the most informative remaining features. Each internal node displays the feature name, threshold used for the split, Gini impurity, the number of samples reaching that node, and the class distribution at that point.

Leaf nodes predicting Returned = 1 tend to be associated with combinations of high-risk signals, such as high prior return counts, size-fit issues, low product ratings, or long delivery times. Leaf nodes predicting Not Returned reflect orders without these elevated risk indicators.

3.3. *Model Configuration*

Parameter	Value / Rationale
max_depth	4: limits tree depth to prevent overfitting while retaining predictive power
random_state	42: ensures reproducible results across runs
criterion	Gini (default): measures node impurity for splitting decisions
Tree depth	4 levels
Leaf nodes	15 (original tree) / fewer with min_samples_leaf=5 (controlled tree)
Processed features	46 (after one-hot encoding of 9 categorical features)

4. **Model Evaluation**

4.1. *Confusion Matrix*

The confusion matrix below shows the breakdown of predictions on the test set (90 records). The rows represent the actual class, whether the order was actually returned. The columns represent the predicted class, what the model said would happen.

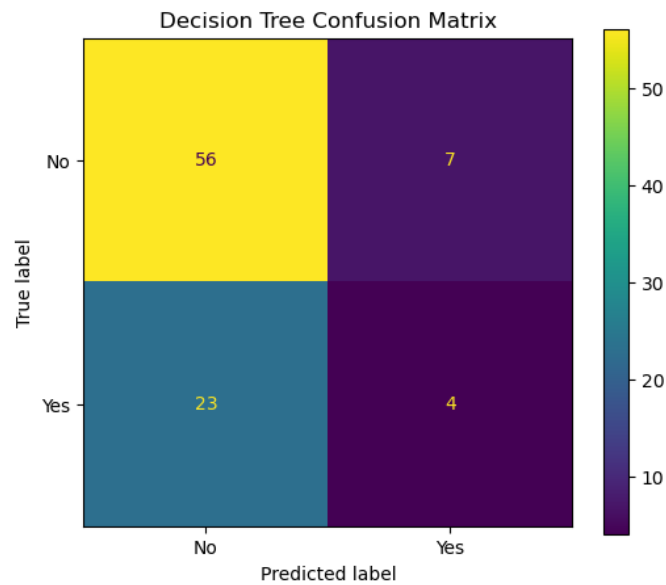


Figure 2: Confusion Matrix, ReturnWise Retail Analytics (*max_depth=4* model)

Reading the matrix:

- **True Negatives (56):** The model correctly predicted 56 orders would not be returned. These were rightly excluded from any intervention, avoiding unnecessary cost and customer disruption.
- **False Positives (7):** The model predicted 7 orders would be returned, but they were not. These represent unnecessary interventions, quality checks or outreach applied to orders that would have resolved without issue.
- **False Negatives (23):** The model predicted 23 orders would not be returned, but they actually were. These are missed prevention opportunities, orders that incurred full reverse-logistics cost without any early warning.
- **True Positives (4):** The model correctly identified only 4 orders that were genuinely returned. These represent the successful, high-value predictions that enabled proactive action.

A false positive occurs when the model predicts an order *will* be returned, but the customer keeps it. In this test set there were 7 false positives. In business terms, the company

sends a proactive message, adds a quality check, or assigns customer support to an order that was never going to be a problem. This wastes staff time and can create a slightly intrusive customer experience if the intervention is visible (e.g. unsolicited outreach). However, false positives are the *less costly* error here, a well-intentioned extra check rarely damages the customer relationship.

A false negative occurs when the model predicts an order *will not* be returned, but the customer does return it. There were 23 false negatives in the test set, a far larger number than false positives. This is the more damaging error. Every missed return means the business incurs the full cost: reverse shipping, inspection, restocking, refund processing, and warehouse labour, all without any early warning. With 23 missed returns out of 27 actual ones, the model is currently catching only 4 returns, leaving the business largely reactive rather than proactive. Reducing false negatives (improving recall) should be the primary model improvement goal.

4.2. *Classification Metrics*

The model achieved a test accuracy of 66.7%, meaning it correctly classified about two-thirds of orders. On its own, this sounds reasonable, but the breakdown by class tells a more important story. For the "No return" class the model performed well (precision 0.71, recall 0.89), but for the "Yes returned" class, the one the business actually cares about, performance was poor: precision of only 0.36 and recall of just 0.15. In plain terms, the model correctly identified only 4 out of 27 actual returns in the test set. For a business whose entire goal is to proactively flag at-risk orders, this is a significant shortfall.

Metric	Result	Business Meaning
Training Accuracy	78.3%	High, expected on data the model was trained on
Testing Accuracy	66.7%	2 in 3 unseen orders correctly classified
Train / Test Gap	~11.6 pts	Mild overfitting, model generalises reasonably
Precision (Yes)	36.4%	Of orders flagged as returns, 36% actually were returned
Recall (Yes)	14.8%	Model identified only 15% of all actual returns, the critical weakness
F1-Score (Yes)	21.1%	Balanced score, the priority metric
False Positives	7	Orders incorrectly flagged as returns, unnecessary interventions

False Negatives	23	Real returns missed, full reverse-logistics cost incurred
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4.3. *The Important Evaluation Metric*

Recall and F1-score are the most important metrics here, not accuracy. Accuracy alone is misleading for this dataset because the classes are imbalanced 63 'No' versus 27 'Yes' in the test set. A naive model that always predicted 'No' would achieve 70% accuracy while being entirely useless for preventing returns.

Recall measures what the business actually needs: of all actual returns, how many did the model identify? At 14.8%, the model currently catches fewer than 1 in 6 actual returns. F1-Score balances this against Precision, ensuring the model does not flood operations staff with false alerts. Both errors carry real cost: missing returns (false negatives) generates direct reverse-logistics and refund expense, while over-flagging safe orders (false positives) wastes staff time and risks intruding on customers unnecessarily.

4.4. *Overfitting Analysis*

The unconstrained Decision Tree (max_depth=None) achieves 100% training accuracy but only 60.0% on the test set, a gap of approximately 40 percentage points. This is severe overfitting: the tree grew without limit and effectively memorised the 360 training examples rather than learning generalizable return patterns. The depth-sweep experiment confirms this clearly, as depth increases, training accuracy climbs steadily while test accuracy peaks and then declines:

Model	Training Accuracy	Testing Accuracy
Controlled (max_depth=4)	78.3%	66.7%
Overfitted (no max_depth limit)	100%	60.0%

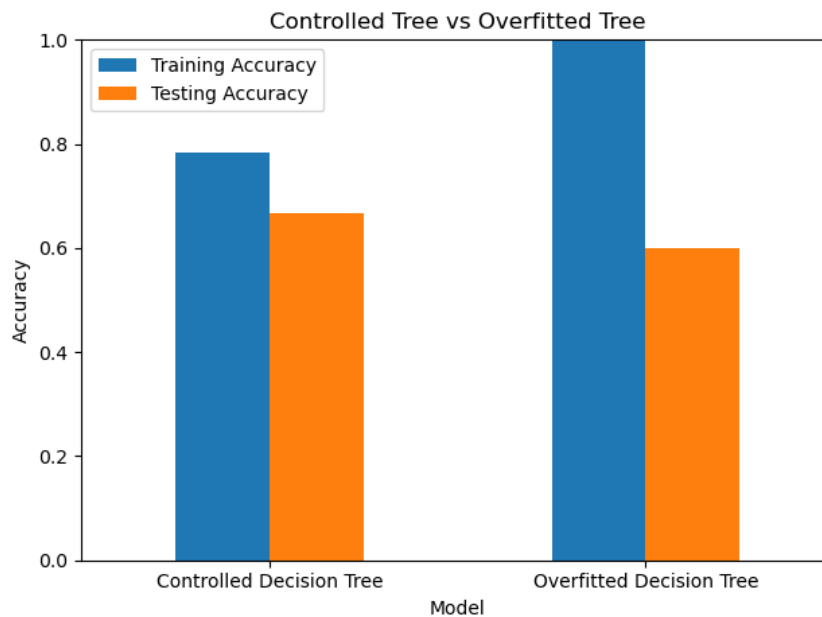


Figure 3: Controlled Tree vs Overfitted Tree, ReturnWise Retail Analytics

5. Controlling Decision Tree Complexity

5.1. Training Accuracy vs Testing Accuracy

There is a meaningful gap between training accuracy and testing accuracy. That is an 11.6 percentage point drop from training to testing. While not catastrophic, this gap is large enough to be concerning and suggests the model is not generalising as well as it could to unseen data.

Comparison	Unconstrained	Original (depth=4)	Controlled (depth=4, leaf=5)
Training Accuracy	100.0%	78.3%	77.2%
Testing Accuracy	60.0%	66.7%	66.7%
Train/Test Gap	~40 pts	~11.6 pts	~10.6 pts
Tree Depth	Unconstrained	4	4
Leaf Nodes	Many	15	Fewer (min_samples_leaf=5)

6. Feature Importance

Feature importance measures how much each feature contributed to reducing Gini impurity across all splits in the Decision Tree. Features with higher scores were used more

frequently and at higher levels of the tree, making them the primary drivers of the model's predictions. The most important feature identified was `Size_Fit_Issue_Level_High` with an importance score of approximately 13.9%, the single strongest predictor of a return.

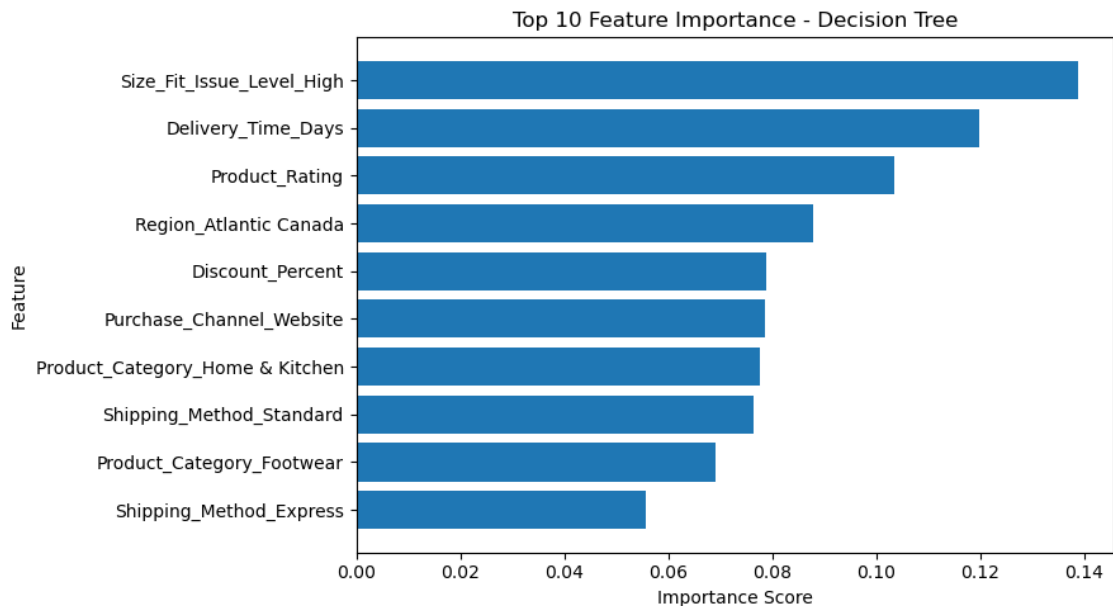


Figure 4: Top Important Features, ReturnWise Retail Analytics

6.1. Business Value of the Most Important Features

Rank	Feature	Importance	Business Interpretation
1	Size_Fit_Issue_Level_High	13.9%	The dominant predictor. When a customer flags a high size or fit concern, the return is almost inevitable. This points directly to gaps in sizing guides, product photography, and measurement information, particularly for apparel and footwear.
2	Delivery_Time_Days	12.0%	The second strongest signal. Longer delivery times erode customer satisfaction and increase the likelihood of a return, especially when expectations were not set clearly at checkout. Fast delivery reduces buyer's remorse and commitment risk.
3	Product_Rating	10.4%	Low-rated products return at significantly higher rates. This is a quality and description signal either the product does not match its listing, or supplier quality is inconsistent. Low-rated SKUs should trigger immediate review.
4	Region_Atlantic_Canada	8.8%	Customers from Atlantic Canada show a distinct return pattern compared to other regions. This may reflect longer delivery times to that region, a narrower product

			range available, or unmet expectations around shipping costs and timelines.
5	Discount_Percent	7.9%	Heavily discounted orders attract price-driven buyers who are more likely to return items after purchase. Deep discounts may reduce perceived product value or attract customers with lower purchase commitment.
6	Purchase_Channel_Website	7.9%	Orders placed through the website carry a different return profile than those placed through other channels (app, marketplace, social). Website buyers may have less guided purchase experiences, leading to more mismatched expectations.
7	Product_Category_Home & Kitchen	7.8%	Home & Kitchen products return at a notable rate, likely due to sizing, compatibility, or aesthetic mismatches that are difficult to assess from product images alone. Better lifestyle photography and specification detail would help.
8	Shipping_Method_Standard	7.7%	Standard shipping is associated with higher return risk, likely because longer wait times increase dissatisfaction and buyer uncertainty before the item even arrives. This connects closely to Delivery_Time_Days.
9	Product_Category_Footwear	6.9%	Footwear has an inherently high return rate due to sizing variability across brands and the difficulty of assessing fit online. Detailed size guides, customer fit reviews, and virtual try-on tools would directly reduce this.
10	Shipping_Method_Express	5.6%	Express shipping also appears in the top 10, but with a different directional role customer who paid for express may have higher expectations and be quicker to return if the product does not meet them.

7. Business Actions and Recommendations

Model Insight	Recommended Business Action
Size_Fit_Issue_Level_High is the top predictor (13.9%)	Invest in detailed sizing guides, fit videos, and virtual try-on tools particularly for Apparel and Footwear. Ensure every product page includes measurement tables, brand-specific size conversion charts, and verified customer fit reviews. This is the single highest-leverage intervention available to the business.

Delivery_Time_Days is the second strongest predictor (12.0%)	Set clear, accurate delivery timeline expectations at checkout rather than generic estimates. Introduce proactive delay notifications before customers begin to wonder. Review carrier performance data by route and region particularly for Standard shipping and consider renegotiating SLAs with underperforming carriers.
Product_Rating correlates strongly with returns (10.4%)	Set an automatic quality review trigger for any product rated below a defined threshold (e.g. 3.5 stars). Update product descriptions, imagery, and supplier agreements for flagged SKUs. Repeat low-rated products should be escalated for removal or supplier renegotiation before they generate further return costs.
Region_Atlantic Canada shows a distinct return pattern (8.8%)	Investigate the root cause before acting on this signal longer transit times, limited product availability, or unmet delivery cost expectations may be driving the pattern rather than customer behaviour. Do not use this feature to deprioritize Atlantic Canada customers. Instead, audit fulfilment performance in that region and improve delivery communication and carrier options where possible.
Discount_Percent correlates with returns (7.9%)	Apply additional product guidance or a personalised confirmation message for heavily discounted orders. Consider flagging high-discount orders for a pre-shipment review. Review whether deep discount promotions are attracting low-commitment buyers and adjust promotional strategy accordingly particularly for high-return categories like Footwear and Home & Kitchen.
Purchase_Channel_Website carries elevated return risk (7.9%)	Audit the website purchase experience for gaps in product guidance. Website buyers may receive less assisted or curated product selection than app or marketplace buyers. Improve on-site filtering, product comparison tools, size guidance pop-ups, and post-purchase confirmation communications to reduce mismatched expectations.
Product_Category_Home & Kitchen drives returns (7.8%)	Conduct a product description audit for the Home & Kitchen category. Returns likely reflect sizing, compatibility, or aesthetic mismatches that are difficult to assess from standard product images. Invest in lifestyle photography, specification tables, and customer Q&A sections. Consider category-specific return window policies where commercially appropriate.
Shipping_Method_Standard is associated with higher return risk (7.7%)	Standard shipping customers wait longer and are more likely to experience buyer uncertainty before delivery. Improve the post-purchase experience for standard shipping orders with proactive tracking updates and estimated arrival confirmations. Consider setting more conservative delivery estimates for standard orders to reduce unmet expectations.
Product_Category_Footwear is a high-return category (6.9%)	Footwear has inherent sizing variability across brands. Implement brand-specific size guides, detailed width and fit descriptors, and customer-verified size reviews on every footwear product page. Explore virtual try-on or fit recommendation tools. Footwear customers who flag a size

	concern (Size_Fit_Issue_Level_High) represent the highest-risk combination in the entire model.
Shipping_Method_Express carries a distinct return signal (5.6%)	Express shipping customers have paid a premium and arrive with higher expectations. If the product does not meet those expectations, they are quicker to act. Prioritise quality checks for express orders and ensure product descriptions are highly accurate for SKUs frequently ordered on express delivery. Consider a brief satisfaction follow-up message shortly after express delivery.
Recall is 14.8% and F1-Score is 21.1%	The model currently catches fewer than 1 in 6 actual returns. It should be used as a risk ranking tool to prioritise human review of the highest-scored orders, not as an automated intervention trigger. Investigate class weighting, SMOTE oversampling for the minority return class, ensemble methods such as Random Forest or Gradient Boosting, and expanding the dataset with real order records to meaningfully improve Recall and F1-Score.

8. Limitations

8.1. Dataset

- **Synthetic dataset:** The dataset was synthetically generated for teaching purposes. Real order data contains entry errors, inconsistent field definitions, seasonal variation, and more complex return patterns than a synthetic sample can replicate.
- **Small sample size:** 450 records is insufficient for a production-grade machine learning model. Each misclassification in the 90-record test set shifts accuracy by approximately 1.1 percentage points, making results sensitive to the random split. A minimum of several thousand records per product category would be needed for reliable deployment.

8.2. Model

- **Very low recall on the return class:** At 14.8%, the model misses 23 of 27 actual returns in the test set. The class imbalance (72% No / 28% Yes) means the tree learns to favour the majority class. Techniques such as class weighting, SMOTE oversampling, or threshold adjustment via `predict_proba` should be explored.

- **Region and Customer_Segment demographic risk:** If certain regions or customer segments historically received poorer product descriptions or slower delivery, the model may learn to over-flag those groups without the root cause being customer behaviour. A fairness audit is required before deployment.
- **Purchase_Date excluded:** Time-based patterns, seasonal return spikes, post-holiday behaviour, promotional event effects were not captured. Incorporating temporal features derived from Purchase_Date could meaningfully improve model performance.

9. Responsible AI Considerations

- **Human review before deployment:** Operations managers should review the model's highest-risk flagged orders before any intervention is applied. The model is a decision-support tool, it should inform, not replace, human judgement about customer context, product quality, and fulfilment circumstances.
- **Fairness and bias audit:** Before any real-world use, the model must be tested for disparate impact across Customer_Segment, Region, and Product_Category. If certain groups historically had higher return rates due to fulfilment failures rather than customer behaviour, the model may have learned to unfairly target them.
- **Explainability:** The Decision Tree structure supports full explainability. The decision path for any individual order prediction can be extracted and explained to managers and customers in plain language, a meaningful advantage over black-box models in a consumer-facing context.
- **Model drift:** Return behaviour changes over time as product ranges, seasonal demand, supplier quality, and customer expectations evolve. The model should be retrained at least quarterly on fresh order data to maintain accuracy and relevance.

- **Data Consent and Retention:** Customer behavioural data must be collected and stored in compliance with applicable privacy legislation, including PIPEDA in Canada, before being used for predictive modelling or targeted operational interventions.